

# **L2b:** Yield, Duration, Convexity, and the Yield Curve

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CHEME 5660 Cornell University

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# Scenario

It is 6:30 AM at your trading desk. You are responsible for a \$500 million portfolio of Treasury securities. The Federal Reserve meets today, and the rumor is a half-point move in the policy rate. The phone is already ringing.

By 2:00 PM you need an answer to one question: how much does this portfolio move?

That question has a conditional answer: it depends on the portfolio cash flows and on which yields move. Today we use the slope and curvature of the price-yield relationship to estimate the change.

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**Careful:** a half-point move in the *policy rate* is not a half-point move in *your security's yield*. We return to that.

# Objectives for Today

Today we take the fixed-income price model from L2a and ask how its price responds to a change in the rate used to discount it.

## Today's concepts

- **Price versus yield**: explain why the price of a fixed positive cash-flow stream decreases with yield and follows a convex curve.
- **Duration and convexity**: define both measures, estimate the local price effect of a yield change, and interpret their units.
- **Yield-curve vocabulary**: distinguish yield to maturity, par yield, spot rates, and the discount factors that price individual cash flows.

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Companion notebook: [L2b lecture notebook](#)

# Lectures and Examples

**Lecture:** CHEME-5660-L2b-Lecture-Yield-Duration-Convexity-Curve-Fall-2026.ipynb

The required example prices a coupon Treasury security and studies how its price responds to a change in yield and to a change in the coupon rate.

**Example:** CHEME-5660-L2b-Example-Treasury-Duration-Convexity-Fall-2026.ipynb

The longer theorem exercise and the term-structure notebooks stay available as optional extensions under advanced/.

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**Link:** <https://github.com/varnerlab/CHEME-5660-CourseRepository-Fall-2026.git>

Quick review before we start: a safe payment can still have a risky price.

# Concept Review: Are Treasury Securities Really Risk-Free?

**No.** Treasuries carry low default risk because they are obligations of the U.S. government, which is why the rate you declare is so often taken from them. That label is conditional. Two risks that remain matter today.

- **Default risk:** the issuer fails to pay. The United States has never repudiated its debt, though a 1979 operational failure did delay payment on some maturing bills. Other sovereigns have defaulted outright: Greece restructured its privately held debt in 2012, Russia defaulted in 1998.
- **Interest-rate risk:** if required yields change, the secondary-market value of a fixed promised cash flow changes. An institution that marks positions to market is affected even without a sale.

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**Today:** interest-rate risk is the one we can compute. Its size depends on maturity, coupon, yield, and the shock.

# Case Study: Interest Rate Risk and Silicon Valley Bank

SVB held a large portfolio of **long-maturity fixed-rate** government securities bought while rates were low. As the Federal Reserve raised rates through 2022 and 2023, the value of those securities fell.

- The losses were unrealized, so a hold-to-maturity story obscured them.
- In early March 2023 uninsured depositors withdrew funds. SVB announced a capital raise to cover realized bond losses, and the bank failed.
- The FDIC took control on March 10, 2023, with roughly \$209 billion in assets: then the second largest failure of an insured bank on record, now the third.

Falling bond values and deposit withdrawals reinforced each other. Interest-rate risk was important, but it was not the only cause of the failure.

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**Overview:** [the collapse of Silicon Valley Bank, from PBS](#)



# Malkiel's Bond-Pricing Theorems

The lecture's destination is a set of classical results, stated now and proved as the tools become available. Each concerns an **option-free** security with fixed positive cash flows, holds the remaining contract terms fixed, and makes its maturity comparisons between bonds priced **at par** ( $c = y$ ).

1. Bond prices move **inversely** to bond yields.
2. **Longer maturity** means more sensitivity to a yield change.
3. The sensitivity gain from added maturity **diminishes** as maturity lengthens.
4. Equal yield moves produce **asymmetric** price changes: a decrease raises the price by more than an equal increase lowers it.
5. **Higher coupon rates** mean less sensitivity, in percentage terms.

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**Reference:** Malkiel (1962), *Expectations, Bond Prices, and the Term Structure of Interest Rates*, QJE 76(2), 197–218.

# Zero-Coupon Treasury Bill

Start with the simplest security: a bill that pays nothing until maturity and then returns its face value. This simple cash-flow pattern makes a bill a useful starting point.

Let  $V_P > 0$  be the face value,  $T > 0$  the remaining maturity in years,  $n$  the number of compounding intervals per year, and  $y$  the nominal annual yield used to discount the single promised payment. The per-period factor  $1 + y/n$  is the growth factor over one interval, so the L2a discount factor is  $\mathcal{D}_{T,0}^{-1}(y; n) = (1 + y/n)^{-nT}$ . The price is given by:

$$V_B = \underbrace{(1 + y/n)^{-nT}}_{\text{discount}} V_P.$$

**Standing assumption:**  $y > 0$  throughout this lecture, as it has been for the Treasury data we use. We will identify every result that depends on this assumption.

# Deriving the Price Sensitivity

Three quantities could move the price: the yield  $y$ , the annual coupon rate  $c$ , and the maturity  $T$ . A bill's contract fixes  $c = 0$ , so  $dc = 0$ . Holding  $V_P$  and  $n$  fixed, only changes in  $y$  and  $T$  remain in the total differential:

$$dV_B = \underbrace{-V_P (1 + y/n)^{-nT} \left( \frac{T}{1 + y/n} \right) dy}_{V_{B,y}} - \underbrace{V_P (1 + y/n)^{-nT} (n \log(1 + y/n)) dT}_{V_{B,T}}.$$

**Why there is no  $dc$  term:** the bill's coupon rate is fixed and cannot change. In the coupon-bearing family of the next section,  $\partial V_B / \partial c = (V_P / n) \sum_j \mathcal{D}_{j,0}^{-1}(y; n) > 0$ , which stays strictly positive at  $c = 0$ . The derivative is not zero; the contract simply requires  $dc = 0$ .

# What Does a Change in Maturity Mean?

A bill promises one payment, so its price depends on  $T$  only through the exponent  $nT$ . No intermediate payment can be added or removed, so  $\partial V_B / \partial T$  is well defined here. But  $dT$  can describe two different comparisons:

- **Changing the contractual maturity:** comparing a 6-month bill with a 12-month bill compares two securities issued on different terms. Smooth for a bill, because only the exponent changes. Not smooth for a coupon security, where it adds or removes scheduled payments.
- **The passage of time:** holding one bill, calendar time  $t$  advances while remaining maturity falls, so  $dT = -dt$  along the position. Smooth for a bill over the whole holding period. For a coupon security, smooth only between coupon dates.

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**Later:** both distinctions return when we price coupon-bearing notes and bonds.

# Yield and Maturity Sensitivities

Read  $V_{B,y}$  and  $V_{B,T}$  as **magnitudes**, not the derivatives themselves: the derivatives are their negatives, which is why the minus signs appear above. Under the standing assumption  $y > 0$ , both are positive:

$$V_{B,y} = \underbrace{V_P(1 + y/n)^{-nT}}_{=V_B} \underbrace{\left( \frac{T}{1 + y/n} \right)}_{>0}, \quad V_{B,T} = \underbrace{V_P(1 + y/n)^{-nT}}_{=V_B} \underbrace{(n \log(1 + y/n))}_{=g_y},$$

where  $g_y := n \log(1 + y/n)$  is the continuously compounded equivalent of  $y$  from L2a. Given a well-posed price,  $1 + y/n > 0$ , the first magnitude is positive for any  $T > 0$ . The second needs  $y > 0$ :  $g_y$  carries the sign of  $y$ , so at  $y = 0$  maturity stops mattering, and for  $-n < y < 0$  a longer bill would be worth *more* and trade above par.

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**Careful:** negative nominal yields have occurred in other sovereign markets. State the condition.

# How Yield and Maturity Change a Bill's Price

## Decreasing in the yield

Holding maturity fixed, a higher yield discounts the face value more heavily, so  $V_B \downarrow$  as  $y \uparrow$ . Buy a 52-week bill at auction; if the yield moves from 2% to 3%, the secondary-market price falls below what you paid.

## Decreasing in the time to maturity

Holding the yield fixed at a positive value,  $V_B \downarrow$  as  $T \uparrow$ . But while you *hold* the bill, calendar time advances and the remaining maturity falls, so  $dT < 0$ : six months in, with the yield unchanged, the price is *higher* than you paid, because fewer periods remain to discount. At  $y = 0$  exactly, every discount factor is one and the price does not move at all.

# Optional: Bill Maturity Sensitivity

The required lecture now moves to coupon-bearing notes and bonds. The advanced bill notebook stays here in the sequence because it extends the maturity discussion we just completed.

**Question:** How does a zero-coupon bill's price sensitivity change as its remaining maturity changes?

**Notebook:** advanced/malkiel/, the Treasury bill maturity example.

# Coupon-Bearing Treasury Notes and Bonds

Now let the security pay coupons. Let  $c$  be the annual coupon rate, so each payment is  $C := (c/n)V_P$ , and let  $N := nT$  count the remaining payment dates, indexed by  $j = 1, \dots, N$ . From L2a:

$$V_B = \mathcal{D}_{N,0}^{-1}(y; n) V_P + C \sum_{j=1}^N \mathcal{D}_{j,0}^{-1}(y; n), \quad \mathcal{D}_{j,0}^{-1}(y; n) = (1 + y/n)^{-j}.$$

Holding that schedule fixed, the yield and coupon sensitivities are:

$$\frac{\partial V_B}{\partial y} = -\frac{1}{n(1 + y/n)} \left[ NV_P \mathcal{D}_{N,0}^{-1}(y; n) + C \sum_{j=1}^N j \mathcal{D}_{j,0}^{-1}(y; n) \right] < 0,$$

$$\frac{\partial V_B}{\partial c} = \frac{V_P}{n} \sum_{j=1}^N \mathcal{D}_{j,0}^{-1}(y; n) > 0.$$

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**Careful:**  $N$  is a discrete schedule index. Differentiating a finite sum whose upper limit is  $nT$  with respect to a smooth  $T$  is not a valid total differential.



# How Bond Features Change Its Price

- **Decreasing in the yield:** now *both* the coupons and the final principal are discounted more heavily, so  $V_B \downarrow$  as  $y \uparrow$ .
- **Increasing in the coupon rate, but not proportional to it:**  $\partial V_B / \partial c$  does not depend on  $c$ , and the discounted par payment survives even at  $c = 0$ . Doubling the coupon doubles the coupon stream, not the price.
- **Maturity does not set the price direction by itself:** a **discount bond** ( $c < y$ ) drifts up toward par as maturity shortens, a **premium bond** ( $c > y$ ) drifts down toward par, and a **par bond** ( $c = y$ ) sits at par on a coupon date.

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**Careful:** Malkiel's maturity result is about *sensitivity to a yield change*, not about the price level.

# Duration and Convexity

Derivatives tell us the direction of a price change. Risk management also needs its size in units that can be compared across positions.

We assume an **option-free** security: its promised cash flows do not change when yields change. Conventional noncallable Treasuries qualify; callable, inflation-indexed, and mortgage-backed securities do not.

Let  $N$  be the number of remaining periods and  $X_j \geq 0$  the dollar cash flow at period  $j$ , with at least one  $X_j > 0$ . The same price formula covers both bills and coupon-bearing securities:

$$V_B(y) = \sum_{j=1}^N X_j \mathcal{D}_{j,0}^{-1}(y; n).$$

A note pays  $X_j = C$  for  $j < N$  and  $X_N = C + V_P$ ; a bill pays  $X_j = 0$  for  $j < N$  and  $X_N = V_P$ .

# Two Derivatives, One Shape

Differentiate the price twice with respect to the yield, holding the cash flows  $X_j$  and the schedule length  $N$  fixed:

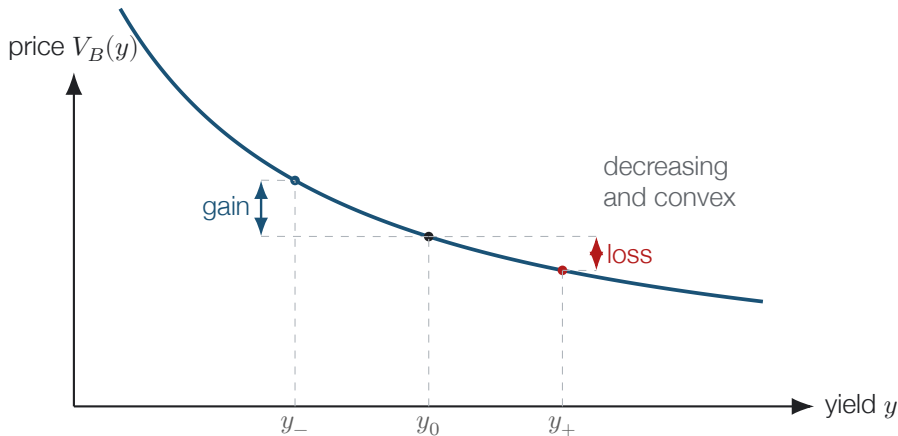
$$\frac{dV_B}{dy} = -\frac{1}{n} \sum_{j=1}^N j X_j (1 + y/n)^{-j-1} < 0,$$

$$\frac{d^2V_B}{dy^2} = \frac{1}{n^2} \sum_{j=1}^N j(j+1) X_j (1 + y/n)^{-j-2} > 0.$$

The strict signs follow from  $1 + y/n > 0$  and from at least one  $X_j$  being strictly positive: no term of either sum is negative, and at least one term of each is positive.

So the price-yield curve is **decreasing** and **strictly convex**. That second fact is called **positive convexity**, and it is what makes the price response asymmetric.

# The Price-Yield Curve



Equal yield moves  $y_{\pm} := y_0 \pm \Delta y$ , for  $\Delta y > 0$ , from the same starting yield  $y_0$ . Convexity makes the **gain** exceed the **loss**.

# The Two Risk Measures

Dividing each derivative by the price removes the size of the position and leaves two local risk measures. With  $V_B(y) > 0$  the price at yield  $y$ , the **modified duration** and the **convexity** are given by:

$$D_{\text{mod}}(y) := -\frac{1}{V_B(y)} \frac{dV_B}{dy}, \quad K(y) := \frac{1}{V_B(y)} \frac{d^2V_B}{dy^2}.$$

Both are positive by the signs just established. With  $y$  a decimal per year,  $D_{\text{mod}}$  has units of years and  $K$  units of years squared.

**Modified is not Macaulay.** The **Macaulay duration**  $D_{\text{mac}}$  is the present-value-weighted average time to payment, and  $D_{\text{mod}} = D_{\text{mac}}/(1 + y/n)$ . For a bill, every dollar arrives at maturity, so  $D_{\text{mac}} = T$  and  $D_{\text{mod}} = T/(1 + y/n)$ , which is exactly the ratio  $V_{B,y}/V_B$  we derived earlier.

# Duration Gives the Slope; Convexity Adds Curvature

A second-order Taylor expansion of  $V_B(y)$  about the current yield, divided by the price, turns the two measures into a repricing estimate. For a yield change  $\Delta y$ , with  $\Delta V_B := V_B(y + \Delta y) - V_B(y)$ , the approximate fractional price change is given by:

$$\frac{\Delta V_B}{V_B} \approx \underbrace{-D_{\text{mod}} \Delta y}_{\text{slope}} + \underbrace{\frac{1}{2}K(\Delta y)^2}_{\text{curvature}}.$$

Because  $K > 0$ , the curvature term is positive for a shock of *either* sign: it adds to the gain and subtracts from the loss.

**The convention.**  $\Delta y$  is a decimal change in the annual yield. One **basis point** is  $10^{-4}$ , so a 50-basis-point shock is  $\Delta y = 0.0050$ .

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**Careful:** a 50-basis-point move in the *policy rate* does not imply  $\Delta y = 0.0050$  for a given security.

# Estimate a Bond's Price Change

Take a note with  $D_{\text{mod}} = 7.2$  years and  $K = 65$  years<sup>2</sup>, and shock its own yield by +50 basis points, so  $\Delta y = 0.0050$ :

- slope term:  $-7.2 \times 0.0050 = -0.0360$
- curvature term:  $\frac{1}{2} \times 65 \times (0.0050)^2 = +0.0008$
- estimate: a **-3.52%** price change, where the straight line alone would have said -3.60%

Both measures are **local**: they are derivatives at the current yield, so the estimate degrades as  $\Delta y$  grows. For the multiple-percentage-point moves of 2022 and 2023 that hurt SVB, the honest calculation is to reprice the whole schedule at the new yield, not to extrapolate from duration.

# What Duration Is Used For

Beyond the single-security estimate, duration is useful for two reasons:

- **A common scale:** securities with different coupons, maturities, and prices land on one scale, so the interest-rate risk of a 2-year note and a 10-year note can be compared directly.
- **A portfolio measure:** the duration of a portfolio is the market-value-weighted average of the durations of the securities it holds, so the \$500 million portfolio from the scenario reprices in one step rather than security by security.

The same properties support risk management: offsetting positions can bring a portfolio's net duration near zero, while a bank that funds long-duration securities with short-duration deposits, as SVB did, carries a duration mismatch.



# Revisiting Malkiel's Theorems

The machinery is in place, so return to the five theorems stated at the start of the lecture. Recall the ground rules: option-free securities with fixed positive cash flows, other contract terms held fixed, and maturity comparisons between bonds **selling at par**,  $c = y$ . That restriction is not a technicality.

Substituting  $C = (c/n)V_P$  and summing the geometric series puts the coupon bond in closed form. Differentiating that at fixed  $c$ , then setting  $c = y$ , gives the modified duration of a par bond:

$$\frac{V_B}{V_P} = \frac{c}{y} + \left(1 - \frac{c}{y}\right) (1 + y/n)^{-N} \quad \Rightarrow \quad D_{\text{mod}}(N) = \frac{1 - (1 + y/n)^{-N}}{y}.$$

**Order matters.** Differentiate at fixed  $c$ , and only *then* set  $c = y$ . Setting  $c = y$  first differentiates along the par locus, where the price is pinned at  $V_P$ , and returns zero. Assume  $y > 0$  throughout.

# Malkiel's Theorems 1, 2, and 3

- **Theorem 1**, prices move inversely to yields: this is the sign  $dV_B/dy < 0$ , or equivalently  $D_{\text{mod}} > 0$ . Every promised payment is positive, so raising  $y$  discounts each more heavily.
- **Theorem 2**, longer maturity increases yield sensitivity:  
 $D_{\text{mod}}(N) = (1 - (1 + y/n)^{-N})/y$  is strictly increasing in  $N$ . A par bond starts at  $V_B = V_P$  at every maturity, so the dollar sensitivity  $-dV_B/dy = V_P D_{\text{mod}}(N)$  rises too: starting at par aligns the percentage and dollar rankings.
- **Theorem 3**, the gain diminishes: one extra period raises modified duration by  $D_{\text{mod}}(N + 1) - D_{\text{mod}}(N) = (1 + y/n)^{-N-1}/n$ , and successive increments shrink by the constant factor  $(1 + y/n)^{-1} < 1$ , so  $D_{\text{mod}}$  climbs toward the ceiling  $1/y$ .

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**Note:**  $1/y$  is the modified duration of a *perpetuity*, a security that pays its coupon forever and never repays principal.

# Malkiel's Theorems 4 and 5

- **Theorem 4**, price movements are asymmetric: a yield decrease raises the price by more than an equal increase lowers it, from the same starting yield. The reason is **positive convexity**. In the second-order estimate the slope term flips sign with the shock while the curvature term  $\frac{1}{2}K(\Delta y)^2$  stays positive. Nonlinearity alone would not do it: a concave curve gives the opposite asymmetry.
- **Theorem 5**, higher-coupon bonds are less sensitive: this is a statement about *percentage* sensitivity, and it needs  $N \geq 2$ . A larger coupon moves present value to earlier dates, lowering the weighted average payment time and so lowering  $D_{\text{mod}}$ . In *dollars* it runs the other way: more cash flow means a larger  $-dV_B/dy$ .

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**Excluded edge:** at  $N = 1$  everything arrives on one date and  $D_{\text{mod}} = 1/[n(1 + y/n)]$ , however large the coupon.

# Why Malkiel Compares Bonds at Par

Outside the par comparison, percentage and dollar sensitivities can rank securities differently. For a zero-coupon bill,  $D_{\text{mod}} = T/(1 + y/n)$  grows without bound, but the dollar sensitivity is a product of a growing and a shrinking factor:

$$-\frac{dV_B}{dy} = V_P \underbrace{\frac{T}{1 + y/n}}_{\text{growing}} \underbrace{(1 + y/n)^{-nT}}_{\text{shrinking}}.$$

So it rises, peaks, and falls. With  $V_P = 100$ ,  $n = 1$ ,  $y = 5\%$  it peaks near  $T \approx 20.5$  years at about 718 USD per unit of yield, then falls to about 541 by  $T = 40$ .

A 40-year bill is far more sensitive than a 20-year bill in **percentage** terms and *less* sensitive in **dollars**. Only the par comparison guarantees the two rankings agree.

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**Example:** CHEME-5660-L2b-Example-Treasury-Duration-Convexity-Fall-2026.ipynb, which tests Theorems 4 and 5.

# From One Yield to the Yield Curve

**Yield to maturity** is the single internal rate that reproduces one security's price from its promised cash flows. It is useful for quoting and comparison, but it is not the full term structure.

A **spot rate** applies to one payment date, so a spot curve supplies one discount factor per date. A **par yield** is the coupon rate that makes a hypothetical security trade at par under that curve; the published Treasury yield curve is a **par yield curve**. Writing  $D(0, t_j)$  for the time-0 discount factor for maturity  $t_j$ , the curve-based price is:

$$V_B = \sum_{j=1}^N X_j D(0, t_j).$$

Curve level, slope, and curvature describe how discounting varies by maturity. They do not, by themselves, identify a unique forecast of future short rates.

# A Recession Predictor?

In most periods long-maturity yields exceed short-maturity yields, so the par yield curve slopes upward. When that ordering flips, the curve is **inverted**.

Inversions have a famous track record: an inverted long-minus-short spread, such as the **10-year minus 3-month** yield, has preceded every U.S. recession of the past half century.

The record comes with qualifiers: false positives occur, lead times run from months to years, and the reading depends on the maturity pair chosen. The optional par-yield example computes the spread from published Treasury data and asks what it is saying now.

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**Notebook:** [advanced/yield\\_curve/CHEME-5660-L3a-ParYieldCurves-Example-Fall-2026.ipynb](#)

# Optional Advanced Material

**Advanced 1:** advanced/malkiel/, the Treasury bill maturity example and the par-bond duration note

**Advanced 2:** advanced/yield\_curve/, a lecture-style notebook with three examples

**Advanced 3:** advanced/strips/, a lecture-style notebook and an issuer example

Optional; these preserve the deeper fixed-income material and are not prerequisites for L3a. The malkiel pair takes Theorem 2 to auction data and proves Theorems 2 and 3 in closed form. The yield-curve sequence covers the investment yield of zeros, yield to maturity by the secant method, and par yield curves from Treasury data, including the recession-indicator spread. The STRIPS sequence infers spot, par, and implied forward rates from STRIPS prices.

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**Notebooks:** [lectures/week-2/L2b/advanced](#), also linked from the lecture notebook.

# Summary

This lecture developed the core tools used to describe and manage the interest-rate risk of fixed-income securities.

## Key takeaways

- **Bond prices decrease as yields increase.** For a fixed stream of positive cash flows, the price-yield curve is decreasing and convex.
- **Duration and convexity measure local price sensitivity.** Duration is the first-order response to a yield change; convexity corrects the straight line for curvature.
- **Malkiel's theorems hold under stated assumptions.** The par restriction is what aligns the percentage and dollar sensitivity rankings.
- **One yield is not a yield curve.** Spot rates value each payment date, and par yields restate that curve as coupon rates.



That's a wrap. What are some of the interesting things we discussed today?